

HCSM-29

The Charge Entropy Observer Triangle

Three Vertices, Three Edges, One Closure

The Hodge Complex Standard Model

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Abstract

The Hodge Complex Standard Model assigns the fourteen modes at the self paired eigenvalue of the discrete Laplacian on the periodic 8×8 torus to one Standard Model generation. Three independent structures organize this multiplet: the charge theorem (HCSM-24), which partitions the modes into three σ -lines by the sign of the zero state amplitude; the orbit decomposition (HCSM-04, HCSM-05), which partitions the modes into three D_4 orbits with equal spaced Shannon entropies; and the observer pair (HCSM-26), the two modes fixed by the chiral reflection rs . This paper proves that these three structures form a triangle in the following precise sense: the observer pair is the unique two element subset of the fourteen mode multiplet that is simultaneously (i) a σ -line, (ii) contained in a single D_4 orbit, and (iii) the fixed locus of the chiral reflection. The triangle closes on the framework constant $d + 1 = 5$, which appears at all three vertices simultaneously: as the charge sum on the observer's σ -line, as the entropy (in bits) of the observer's orbit, and as twice the observer pair's $N/4$ sum. The paper states the triangle theorem, derives the three vertex identities, and gives the closure relation $q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d + 1$. All claims are HCSM-native and verified at machine precision.

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1 Introduction

1.1 Three structures on one multiplet

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1]. The fourteen modes at the self-paired eigenvalue $\lambda = 4$ of the discrete Laplacian are identified with one Standard Model generation. On this multiplet, three independent structures act:

1. **The charge structure** (HCSM-24 [6]). The zero state vector s assigns to each mode a charge-line sign $\sigma = \text{sign}(s) \in \{-1, 0, +1\}$. The three σ -lines partition the fourteen modes into sizes (6, 2, 6) with charge sums (0, +5, -8) in units of $e/3$.
2. **The entropy structure** (HCSM-04, HCSM-05 [2, 3]). The D_4 point group of the square decomposes the fourteen modes into three orbits $\mathcal{O}_0, \mathcal{O}_1, \mathcal{O}_2$ of sizes (2, 8, 4), with Shannon entropies (6.0, 5.5, 5.0) bits and equal spacing 0.5 bits.
3. **The observer structure** (HCSM-26 [7]). The chiral reflection rs of D_4 fixes exactly two modes on the fourteen-mode multiplet: the W boson and the top quark. Their pair $\{W, t\}$ is the observer pair.

1.2 The one-line summary

Three structures charge, entropy, and observer organize the same fourteen modes. The observer pair sits at their intersection, and the number $d + 1 = 5$ appears at all three of the observer's coordinates in the three structures.

1.3 Main result

The paper proves the following triangle theorem.

Charge Entropy Observer Triangle Theorem (Section 3). The observer pair $\{W, t\}$ is the unique two-element subset of the fourteen-mode multiplet that satisfies simultaneously:

- (T1) it is a σ -line of the charge theorem;
- (T2) it is contained in a single D_4 orbit;
- (T3) it is the fixed locus of the chiral reflection rs .

Moreover, the triangle closes on the constant $d + 1$, in the sense that three framework quantities evaluated at the observer's three coordinates coincide:

$$q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d + 1 = 5. \quad (1)$$

1.4 Organization

Section 2 states the three structures. Section 3 proves the triangle theorem. Section 4 derives the three vertex identities. Section 5 derives the closure relation. Section 6 discusses the structural interpretation. Section 7 gives reproducibility notes. Section 8 concludes.

2 Setup: three structures on the fourteen-mode multiplet

All statements in this section are HCSM-native; the reader is referred to HCSM-00 [1] for the full foundations.

2.1 The fourteen-mode multiplet

The substrate is the periodic eight-by-eight torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = 64$ sites. The discrete five-point Laplacian has self paired eigenvalue $\lambda = 4$ with multiplicity 14. In the standard mode order,

$$(\nu, e, u, d, s, \mu, \text{dark}, c, \tau, b, W, Z, H, t). \quad (2)$$

The mode coordinates in this order are:

$$\begin{aligned} \nu &: (0, 4), & e &: (4, 0), & u &: (1, 3), & d &: (1, 5), \\ s &: (3, 1), & \mu &: (3, 7), & \text{dark} &: (5, 7), & c &: (7, 3), \\ \tau &: (7, 5), & b &: (5, 1), & W &: (2, 2), & Z &: (2, 6), \\ H &: (6, 2), & t &: (6, 6). \end{aligned} \quad (3)$$

2.2 The charge structure

The charge theorem of HCSM-24 states that the electric charge $q_3 = 3Q$ (in units of $e/3$) is a bijection

$$F : (n_1, \sigma) \mapsto q_3, \quad n_1 \in \mathbb{Z}_8, \quad \sigma = \text{sign}(s) \in \{-1, 0, +1\}, \quad (4)$$

from the fourteen realized pairs to the five Standard Model charge eigenvalues $\{-3, -1, 0, +2, +3\}$. The three σ -lines are:

$$\sigma = -1 : \{\nu, d, s, \text{dark}, c, Z\}, \quad \text{charge sum } 0, \quad (5)$$

$$\sigma = 0 : \{W, t\}, \quad \text{charge sum } +5, \quad (6)$$

$$\sigma = +1 : \{e, \mu, \tau, b, H, u\}, \quad \text{charge sum } -8. \quad (7)$$

The grand total is $-3 = -1e$.

2.3 The entropy structure

Under the D_4 point group of the square, the fourteen modes decompose into three orbits:

$$\begin{aligned} \mathcal{O}_0 &= \{(0, 4), (4, 0)\}, \quad \text{size } 2, \quad \text{support } 64, \\ \mathcal{O}_1 &= \{(1, 3), (1, 5), (3, 1), (3, 7), (5, 1), (5, 7), (7, 3), (7, 5)\}, \quad \text{size } 8, \quad \text{support } 48, \\ \mathcal{O}_2 &= \{(2, 2), (2, 6), (6, 2), (6, 6)\}, \quad \text{size } 4, \quad \text{support } 32. \end{aligned} \quad (8)$$

The support classes are $\{64, 48, 32\} = \{N, V, N/2\}$ with $V = (d - 1) \cdot 2^d = 48$. The Shannon entropies are

$$H(\mathcal{O}_0) = 6.0 \text{ bits}, \quad H(\mathcal{O}_1) = 5.5 \text{ bits}, \quad H(\mathcal{O}_2) = 5.0 \text{ bits}, \quad (9)$$

with equal spacing 0.5 bits.

In the standard mode order, the orbits are $\mathcal{O}_0 = \{\nu, e\}$, $\mathcal{O}_1 = \{u, d, s, \mu, \text{dark}, c, \tau, b\}$, $\mathcal{O}_2 = \{W, Z, H, t\}$.

2.4 The observer structure

The chiral reflection $rs : (n_1, n_2) \mapsto (n_2, n_1) \pmod{8}$ is the unique element of D_4 that both fixes exactly the pair $\{(2, 2), (6, 6)\}$ and anticommutes with the winding generator H_{wind} [7]. Its fixed set on the fourteen-mode multiplet is the observer pair:

$$\text{Fix}(rs) = \{(2, 2), (6, 6)\} = \{W, t\}. \quad (10)$$

The observer pair's $N/4$ labels are $N/4(W) = 2$ and $N/4(t) = 1/2$, with sum

$$N/4(W) + N/4(t) = 2 + \frac{1}{2} = \frac{5}{2} = \frac{d+1}{2}. \quad (11)$$

3 The triangle theorem

3.1 Statement

Theorem 3.1 (Charge–Entropy–Observer Triangle). *The observer pair $\{W, t\}$ is the unique two-element subset of the fourteen mode multiplet satisfying all three of:*

- (T1) *it is a σ -line of the charge theorem;*
- (T2) *it is contained in a single D_4 orbit;*
- (T3) *it is the fixed locus of the chiral reflection rs .*

3.2 Proof

Proof. Step 1: (T3) selects $\{W, t\}$. By the Fixed-Locus Theorem of HCSM-26 (§3.1), the fixed set of rs on the fourteen-mode multiplet is exactly $\{(2, 2), (6, 6)\} = \{W, t\}$. So (T3) forces the set to be $\{W, t\}$.

Step 2: (T1) is satisfied. By (6), the σ -line at $\sigma = 0$ is exactly $\{W, t\}$. So $\{W, t\}$ satisfies (T1).

Step 3: (T2) is satisfied. Both W and t lie in $\mathcal{O}_2$ (orbit (8)), since $W : (2, 2)$, $Z : (2, 6)$, $H : (6, 2)$, $t : (6, 6)$ form the orbit $\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}$. So $\{W, t\} \subset \mathcal{O}_2$, satisfying (T2).

Step 4: uniqueness. We must show no other two-element subset satisfies all three conditions simultaneously.

- Condition (T3) is the strongest: it uniquely determines the set $\{W, t\}$ (Lemma 3.1 of HCSM-26).
- Therefore any subset satisfying all three must equal $\{W, t\}$.
- To confirm the conditions are compatible, we have verified (T1) and (T2) above.

Hence $\{W, t\}$ is the unique subset satisfying (T1), (T2), and (T3). □

Remark 3.2 (Independence of the conditions). Conditions (T1) and (T2) alone admit many candidates: any pair drawn from a single orbit and lying on a single σ -line. The pair $\{W, t\}$ is the unique one that is *simultaneously* a σ -line (size-2), a subset of a single orbit, and the fixed locus of a chiral involution. Condition (T3) is what forces uniqueness; conditions (T1) and (T2) are what make the set a structurally distinguished vertex.

4 Vertex identities

We now evaluate three framework quantities at the observer's three coordinates (charge, entropy, observer).

4.1 Charge vertex

Proposition 4.1 (Charge vertex). *The charge sum on the observer's σ -line is*

$$q_3(\sigma = 0) := q_3(W) + q_3(t) = (+3) + (+2) = +5 = d + 1. \quad (12)$$

Proof. From HCSM-24 (§10.1), the charge assignments are $q_3(W) = +3$ and $q_3(t) = +2$. Sum: $+5$. At $d = 4$: $d + 1 = 5$. $\square$

4.2 Entropy vertex

Proposition 4.2 (Entropy vertex). *The Shannon entropy (in bits) of the orbit containing the observer pair is*

$$H(\mathcal{O}_2) = 5.0 \text{ bits} = d + 1. \quad (13)$$

Proof. From HCSM-04, the orbit $\mathcal{O}_2$ has entropy class 5.0 bits, and the three orbit entropies are (6.0, 5.5, 5.0) bits with equal spacing. At $d = 4$: $d + 1 = 5$. $\square$

4.3 Observer vertex

Proposition 4.3 (Observer vertex). *Twice the observer pair's $N/4$ sum is*

$$2 \cdot [N/4(W) + N/4(t)] = 2 \cdot \frac{5}{2} = 5 = d + 1. \quad (14)$$

Proof. From HCSM-23 and HCSM-26, $N/4(W) = 2$ and $N/4(t) = 1/2$. Sum: $5/2$. Doubled: $5 = d + 1$. $\square$

Remark 4.4 (The factor of 2). The factor of 2 in (14) is not arbitrary: it is the same factor that appears in the dimensional-rigidity chain of HCSM-26 (§9), where the observer-pair separation $N/4(W) - N/4(t) = (d - 1)/2$ doubles to $d - 1$ and the identity $(d - 1)^2 = 2d + 1$ closes at $d = 4$. The observer vertex natively carries the “half” value $(d + 1)/2$, and doubling recovers the framework constant $d + 1$ common to all three vertices.

5 The closure relation

5.1 The triangle closure

Combining Propositions 4.1, 4.2, and 4.3:

Theorem 5.1 (Triangle closure). *The three vertices of the charge–entropy–observer triangle coincide on the framework constant $d + 1$:*

$$\boxed{q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d + 1 = 5.} \quad (15)$$

5.2 Why the closure works

The three identities in (15) close on $d + 1 = 5$ because of the following structural facts:

- **Charge vertex.** The σ -line at $\sigma = 0$ consists of the pair $\{W, t\}$, whose Standard Model charges are $+3$ (W) and $+2$ (t). Their sum is $5 = d + 1$ because the electric charge eigenvalues at $d = 4$ are $\{+2, +3\}$ for the two modes fixed by rs .

- **Entropy vertex.** The orbit $\mathcal{O}_2$ has entropy class 5.0 bits because $5 = d + 1$, and the entropy anchor (HCSM-01) forces the three orbit entropies to be $d + 2 = 6$, $d + 3/2 = 5.5$, and $d + 1 = 5$.
- **Observer vertex.** The observer pair’s $N/4$ sum is $(d + 1)/2 = 5/2$, and doubling recovers $d + 1$.

5.3 Two independent closures

The closure in (15) can also be stated as two independent pairs of equalities:

Charge–Entropy closure. The charge sum on the observer’s σ -line equals the entropy (in bits) of the observer’s orbit:

$$q_3(\sigma = 0) = H(\mathcal{O}_2). \quad (16)$$

Observer–Sum closure. Twice the observer pair’s $N/4$ sum equals the common value:

$$2 \cdot [N/4(W) + N/4(t)] = q_3(\sigma = 0) = H(\mathcal{O}_2). \quad (17)$$

Both closures are exact at machine precision. Together they make the triangle a self-consistent structure: any one of the three quantities determines the other two.

6 Discussion

6.1 What the triangle is

The charge–entropy–observer triangle is not a new partition of the fourteen-mode multiplet. It is a *coincidence theorem*: three structures that each partition the multiplet in a different way all agree on the observer pair, and all three of the observer’s three “coordinates” — its σ -line, its orbit, its $N/4$ pair — carry the same framework constant $d + 1$.

6.2 Relation to the observer-pair uniqueness theorem

The Observer-Pair Uniqueness Theorem of HCSM-26 (§7.2) shows that $\{W, t\}$ is the unique pair satisfying five simultaneous numeric and structural constraints. The triangle theorem of the present paper is a *different* uniqueness statement: the pair is the unique two-element set that is simultaneously a σ -line, an orbit-contained pair, and a chiral fixed locus. The two theorems are complementary.

6.3 Relation to the dimensional-rigidity chain

The observer-pair uniqueness file records a chain of three identities that all reduce to $d^2 - 4d = 0$:

- Observer-pair separation: $d/2 - 2/d = (d - 1)/2$;
- kL crossover: $(d - 1)^2 = 2d + 1$;
- Dark energy coefficient: $(d + 1)/2 = |F|/|B|$.

The triangle closure (15) is the “counting” partner of this chain: where the rigidity chain exhibits $d = 4$ through three algebraic identities, the triangle exhibits $d + 1 = 5$ through three structural coincidences.

6.4 Why $\sigma = 0$

The charge vertex sits at $\sigma = 0$ because that is the unique σ -line that is invariant under the chiral reflection rs . By the Fixed-Locus Theorem of HCSM-26, rs swaps the $\sigma = \pm 1$ lines and fixes the $\sigma = 0$ line pointwise. So the observer’s charge coordinate is determined by its chiral structure — it must be $\sigma = 0$.

6.5 Why the orbit is $\mathcal{O}_2$

The entropy vertex sits at $\mathcal{O}_2$ because that is the unique orbit containing the fixed points of rs . Both $W = (2, 2)$ and $t = (6, 6)$ lie on the diagonal $n_1 = n_2$, so they must lie in $\mathcal{O}_2 = \{(2, 2), (2, 6), (6, 2), (6, 6)\}$. The other orbits — $\mathcal{O}_0$ and $\mathcal{O}_1$ — contain no diagonal modes.

6.6 Why the factor of 2

The observer vertex carries the “half” value $(d+1)/2$ because the observer pair is a two-element set: the $N/4$ sum of *one* mode is 2, of the *other* is $1/2$, and the sum $5/2$ is exactly half the framework constant $d+1$. Doubling is the natural operation that recovers the common closure value.

7 Reproducibility

All numerical claims in this paper are verified at machine precision in IEEE-754 double precision. The verification script is provided in the HCSM companion repository at `12-VERIFICATION/HCSM-29_verification`. It constructs the fourteen-mode multiplet, the σ -lines, the D_4 orbits, the observer pair, and the $N/4$ labels from HCSM-native definitions, and verifies:

1. The σ -line sizes $(6, 2, 6)$ and the charge sums $(0, +5, -8)$.
2. The orbit sizes $(2, 8, 4)$ and the Shannon entropies $(6.0, 5.5, 5.0)$ bits.
3. The fixed locus of rs equals $\{W, t\}$.
4. The observer pair is contained in $\mathcal{O}_2$.
5. The three vertex identities (12), (13), (14).
6. The closure relation (15).

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

8 Conclusion

This paper has established the charge–entropy–observer triangle of the Hodge Complex Standard Model. The three vertices are the σ -line structure (charge), the D_4 orbit entropy structure, and the observer pair. The observer pair $\{W, t\}$ is the unique two-element subset that is simultaneously a σ -line, contained in a single orbit, and the fixed locus of the chiral reflection rs . The triangle closes on the framework constant $d+1 = 5$, which appears at all three vertices: as the charge sum on the observer’s σ -line, as the entropy (in bits) of the observer’s orbit, and as twice the observer pair’s $N/4$ sum.

The closure relation

$$q_3(\sigma = 0) = H(\mathcal{O}_2) = 2 \cdot [N/4(W) + N/4(t)] = d + 1 = 5 \quad (18)$$

is a numerical identity of the HCSM substrate, verified at machine precision. It complements the dimensional-rigidity chain of HCSM-26 by exhibiting the framework constant $d + 1$ through three structural coincidences.

References

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- [9] Particle Data Group (S. Navas et al.), *Review of Particle Physics*, Phys. Rev. D **110**, 030001 (2024).

A Web Crawler Metadata

A.1 Dublin Core

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subset that is simultaneously a sigma-line, contained in a single orbit, and the fixed locus of the chiral reflection. The triangle closes on the framework constant $d+1 = 5$, which appears at all three vertices: as the charge sum on the observer's sigma-line, as the entropy (in bits) of the observer's orbit, and as twice the observer pair's $N/4$ sum. The closure relation

$$q_3(\sigma=0) = H(O_2) = 2*[N/4(W) + N/4(t)] = d+1 = 5$$

is verified at machine precision.

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A.2 Schema.org JSON-LD

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A.3 BibTeX for Automated Indexing

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